

Lab 07

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1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(RSQLite)
library(DBI)

conn <- dbConnect(RSQLite::SQLite(), "C:/Users/joaoz/Desktop/Lab07/uwmadison.sqlite3")

dbListTables(conn)
```

```
[1] "course_offerings"  "courses"           "grade_distributions"
[4] "instructors"       "rooms"              "schedules"
[7] "sections"          "subject_memberships" "subjects"
[10] "teachings"
```

```
dbListFields(conn, "teachings")
```

```
[1] "instructor_id" "section_uuid"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
dbGetQuery(conn, "SELECT DISTINCT
  instructors.id,
  instructors.name
FROM subjects
JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code
JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid
JOIN sections
```

```

    ON course_offerings.uuid = sections.course_offering_uuid
JOIN teachings
    ON sections.uuid = teachings.section_uuid
JOIN instructors
    ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
ORDER BY instructors.name;")

```

	id	name
1	4787911	ABIGAIL BENZINE
2	3111842	ADIN-CRISTIAN ANDREI
3	3775809	AKICHIKA OZEKI
4	4958796	ALAN HUANG
5	5451706	ALBERTO DEL PIA
6	3661782	ALBRECHT KLEMM
7	5656034	ALEXANDER COVINGTON
8	4680082	ALEXANDER FISH
9	3573775	ALEXANDER KISELEV
10	5162339	ALYSSA DIGILIO
11	129415	ANATOLE BECK
12	3674168	ANDREJ ZLATOS
13	5003565	ANDREW LESLIE
14	4952769	ANNE BRUCKNER
15	5498092	ANQI SHI
16	4646729	ANRU ZHANG
17	2600893	ARNOLD MILLER
18	4963378	AUGUST JENSEN
19	5852265	BEHZAD AALIPUR
20	5694563	BEHZAD AALIPUR HAFSHEJANI
21	4140469	BEN ADAM HAALAND
22	4538519	BENEDEK VALKO
23	4632211	BENJAMIN RECHT
24	5906523	BI CHENG WU
25	4155990	BIN DAI
26	5186047	BIN ZHANG
27	4436066	BIN ZHU
28	5144884	BINGYING XIE
29	3484236	BO HUANG
30	5632570	BO YANG
31	5690247	BOWEN HU
32	4753018	BRET HANLON
33	751075	BRET LARGET

34	2602055	BRIAN YANDELL
35	5011312	BROOK LUERS
36	3955546	CECILE ANE
37	5883206	CHAN PARK
38	3976099	CHANHAN HSU
39	4724058	CHAOQUN MEI
40	4712098	CHAOYANG YU
41	3376532	CHELSEY GREEN
42	4586558	CHEN CHENG
43	5018463	CHEN JING
44	5592111	CHENGNING ZHANG
45	4597046	CHENLIANG XU
46	4903489	CHENSHENG KUANG
47	3973813	CHENXI LI
48	3629214	CHIA-CHIEH LIN
49	3613299	CHIEN-WEI CHEN
50	340544	CHRISTINA M. KENDZIORSKI
51	138086	CHRISTINE SORKNESS
52	3855025	CHRISTOPHER WAGNER
53	5887049	CLAIRE BOBST
54	4721578	CLAUDIA SOLIS LEMUS
55	4312759	COLE COOK
56	5332940	COLIN LONGHURST
57	5498648	CRYSTAL CHEN
58	5015194	CUICUI QI
59	5176022	CUIZE HAN
60	5846795	DANIELE CAPPELLETTI
61	4308515	DAVID ANDERSON
62	2601596	DAVID DEMETS
63	2602109	DAVID GRIFFEATH
64	6043299	DEBRAJ DAS
65	4264265	DEREK BEAN
66	5305675	DEREK NORTON
67	3773167	DEYUAN JIANG
68	2600233	DIETRICH UHLENBROCK
69	393380	DONALD PORTER
70	4898014	DONG XIA
71	4897725	DONGGYU KIM
72	5620719	DONGHYUN LEE
73	2602050	DOUGLAS M. BATES
74	4728844	DUY NGUYEN
75	5516726	DUZHE WANG
76	5189535	EDWARD ERKER

77	2824494	ELOISA D CHAVAS
78	3566378	ERICA LEE DEADMAN
79	2602042	ERIK NORDHEIM
80	5204089	FAN GAO
81	4437355	FAN YANG
82	3460189	FANG FANG
83	3783301	FANGFANG WANG
84	4601948	FLORIAN BERTRAND
85	361804	FREDERICK BOEHM
86	5214164	GARVESH RASKUTTI
87	495087	GARY HOWARD SCHROEDER
88	4957688	GENG LI
89	5017655	GINA BENNINGER
90	5862942	GINA OH
91	5320625	GONZALO CONTADOR
92	2600488	GRACE WAHBA
93	4761200	GREGORIO MORENO-FLORES
94	5127444	GREGORY SHINAULT
95	4272783	GUANHUA CHEN
96	4436002	GUANNAN SUN
97	1505756	GUILHERME ROSA
98	4886028	GUILHERME VIEIRA NUNES LUDWIG
99	4884161	GUN WOONG PARK
100	5010171	HAN CHEN
101	5514514	HAO CHEN
102	5520701	HAO TENG
103	4435790	HAO ZHENG
104	5323690	HAO ZHOU
105	3456935	HAODA FU
106	4725072	HAOYANG FAN
107	4149843	HEATHER MARIE BRAZEAU
108	3623432	HUAIBAO FENG
109	3483534	HUI WANG
110	5168069	HUIKUN ZHANG
111	5167990	HUILIN HU
112	5321537	HYEBIN SONG
113	4734607	HYUNSEUNG KANG
114	801445	IAN BRANSTAD RILEY
115	143932	ISMOR FISCHER
116	5515948	JAMES ANDERSON
117	2601633	JAMES D KUELBS
118	4636315	JARED BROWN
119	3011936	JASON P FINE

120	3956063	JASON RICHARD SWANSON
121	4140397	JEE YEON KIM
122	4890283	JEEA CHOI
123	4636145	JENNIFER BIRSTLER
124	4613728	JENNIFER NGUYEN
125	3627934	JESSE THOMAS HOLZER
126	4590715	JIAJIE CHEN
127	4433317	JIALE XU
128	4603165	JIAN WU
129	4886987	JIANCHANG HU
130	5410648	JIE SONG
131	4292460	JIE WEI
132	4436368	JIE ZHANG
133	4734274	JILI WANG
134	3783649	JINGCI MENG
135	3973318	JINGJIANG PENG
136	4902482	JINGLAN LI
137	4286103	JIWEI ZHAO
138	4880968	JOHN DAVIS
139	608997	JOHN GILLET
140	3806328	JOHN KANE
141	4896511	JOHN WILTSHIRE-GORDON
142	1634418	JONATHON PETERSON
143	1126199	JOSEP GINEBRA
144	4610245	JOSEPH DEUTSCH
145	4731731	JU HEE CHO
146	3981051	JUN LI
147	1079018	JUN SHAO
148	3620114	JUN YIN
149	3781764	JUN ZHANG
150	321374	JUN ZHU
151	3095399	JUNGWON MUN
152	3620552	JUNHEE HAN
153	4541633	JUNHO LEE
154	5012817	KAILEI CHEN
155	2602032	KAM-WAH TSUI
156	4541610	KARL BROMAN
157	4929500	KARL ROHE
158	4491860	KATHERINE GOODE
159	3777992	KAZUHIKO SHINKI
160	4261114	KEEGAN KORTHAUER
161	3127329	KEN ONO
162	3978606	KEVIN HASEGAWA ENG

163	4744350	KEVIN PACKARD
164	3404299	KJELL DOKSUM
165	5476019	KRISHNAKUMAR BALASUBRAMANIAN
166	4199106	KRISTEN CYFFKA
167	5697552	KUNLING HUANG
168	5317373	KYLE HEBERT
169	5478075	KYLE HERBET
170	207204	KYUNGSMANN KIM
171	4435446	LAM HO
172	5521838	LAN LUO
173	3103459	LANCINE KONATE
174	3443667	LEI XU
175	3437432	LEV BORISOV
176	4845798	LIAM JOHNSTON
177	4277002	LIE XIONG
178	3980675	LILI LAN
179	5693114	LILI ZHENG
180	4578330	LILUN DU
181	5514932	LIN QI
182	5015113	LONG PHAN
183	4726678	LU MAO
184	5010560	LU YANG
185	5017774	LUWAN ZHANG
186	5192960	LUXI CAO
187	5302154	MANJUSHA KANCHARLA
188	2601674	MARI PALTA
189	4195000	MARIA KAMENETSKY
190	5167906	MARIA ROJO
191	245914	MARIAN R FISHER
192	333334	MARY LINDSTROM
193	3467530	MATTHEW BALLARD
194	4280220	MENG SONG
195	1114818	MICHAEL FERRIS
196	783769	MICHAEL GEORGE ILTIS
197	270258	MICHAEL HOGAN
198	5708167	MICHAEL KUTZLER
199	5840930	MICHAEL LIOU
200	2601916	MICHAEL NEWTON
201	989414	MICHAEL RENE KOSOROK
202	394810	MICHELLE HARRIS
203	6029366	MIHAELA IFRIM
204	3575433	MIN JUNG LEE
205	3965653	MIN NIU

206	4133622	MING XIE
207	3077702	MING YUAN
208	4392906	MINJING TAO
209	4947463	MITCHELL PAUKNER
210	3404298	MOO K CHUNG
211	5514561	MUHONG GAO
212	2601621	MURRAY CLAYTON
213	1505727	NATALIA DE LEON GATTI
214	232828	NELLIE LAUGHLIN
215	3523865	NICHOLAS HENDERSON
216	640407	NICHOLAS STEPHEN KEULER
217	5523214	NING FAN
218	3882760	NORBERT BINKIEWICZ
219	2600689	NORMAN DRAPER
220	2601093	PAUL M TERWILLIGER
221	967335	PAUL RATHOUZ
222	5631398	PAUL SAVARIAPPAN
223	5326263	PEIGEN ZHOU
224	4149787	PERLA REYES
225	1515839	PHILIP WOOD
226	4633355	QI JIANG
227	3974100	QI TANG
228	3338237	QIAN ZHIGUANG
229	5718599	QING LI
230	4460098	QIONG ZHANG
231	4582107	QIURONG CUI
232	4432994	QUEFENG LI
233	4161984	QUOC TRAN
234	789363	REBECCA KOSCIK
235	5488796	REBECCA POST
236	2600141	RICHARD A BRUALDI
237	2601613	RICHARD A. JOHNSON
238	2601926	RICHARD J. CHAPPELL
239	4957895	ROBERT HARRON
240	2601512	ROBERT R MEYER
241	2601091	ROBERT WARDROP
242	3780010	ROBERT WAYNE GREEN
243	893827	RONALD GANGNON
244	5887858	RONGJUN ZHU
245	5690182	RUI CHEN
246	4128750	RUI TANG
247	3987897	RUIFANG SONG
248	3343055	RUIFENG XU

249	5898424	RUNGANG HAN
250	5020061	RUOSI GUO
251	3727371	RYAN ZEA
252	4752238	SAMUEL STECHMANN
253	3981781	SANGBUM CHOI
254	5354849	SCOTT HOTTOVY
255	3253489	SCOTT JOSEPH HETZEL
256	5068296	SEAN KENT
257	5110390	SEBASTIEN ROCH
258	4733192	SEHO PARK
259	5520811	SEULKEE YUN
260	3573852	SEUNGBONG HAN
261	5905014	SHAN LU
262	982957	SHANE HUBLER
263	5515201	SHANG WU
264	4146030	SHENG WANG
265	5906594	SHENG WANG
266	4285558	SHENG ZHANG
267	5017571	SHENGJI JIA
268	3127330	SHI JIN
269	4885477	SHIXUE LIU
270	4598772	SHIZHEN WANG
271	4699427	SHUAI CHEN
272	4271924	SHUANG HUANG
273	5170885	SHULEI WANG
274	4588354	SHUYUN YE
275	3622809	SIJIAN WANG
276	5323628	SIJING LI
277	5176905	SOHEIL SADEGHI
278	4593965	SOKOL VAKO
279	3955531	SONA Z SWANSON
280	4953246	SONG WANG
281	3884290	SRIKANTHMADHAVAN ARAVAMUTHAN
282	2600973	STEFFEN LEMPP
283	3122658	STEPHEN AARON STANHOPE
284	4538540	STEPHEN BERG
285	2601051	STEPHEN WAINGER
286	3382451	STEPHEN WRIGHT
287	5452538	STEVEN SAM
288	3117835	SUKHENDU MEHROTRA
289	3801422	SUNDUZ KELES
290	4590738	TAERI UHM
291	3781422	TAO YU

292	5023414	THEVAASIINEN CHANDERENG
293	417071	THOMAS COOK
294	2601039	THOMAS G. KURTZ
295	4904255	THU LE
296	4896939	TIEN VO
297	3374880	TIMO SEPPALAINEN
298	4879129	TIMOTHY IDOWU
299	4957916	TIMOTHY IDOWU
300	5520995	TING YE
301	3351080	TING-LI LIN
302	5329574	TONG LI
303	3230527	TONGHAI YANG
304	4896172	TRAM TA
305	5567363	TUN LEE NG
306	5664494	TZU HSIANG HUNG
307	3959251	VARSHA KULKARNI
308	4641200	VICTOR LUO
309	4788841	VICTORIA MANSFIELD
310	4434130	WAI TONG FAN
311	3780817	WEI ZHENG
312	2602108	WEI-YIN LOH
313	4590481	WENWEN ZHANG
314	5322094	WENZHI CAO
315	5161294	WESLEY CHANG
316	4591139	XIAO GUO
317	4905505	XIAO NIE
318	3343005	XIAODAN WEI
319	5170801	XIAOMAO LI
320	4729481	XIAOPING FENG
321	5258581	XIAOWU DAI
322	4285528	XIN LI
323	4588103	XIN ZHANG
324	4726666	XINJIE HE
325	3620612	XINWEI DENG
326	4286679	XINXIN YU
327	4430246	XINYU SONG
328	4145889	XIPEI YANG
329	5170924	XIRAN WANG
330	4194670	XIUFENG SHAO
331	5322566	XIUYU MA
332	3791802	XIWEN MA
333	4286138	XU HE
334	4590559	XU XU

335	4726703	XUEYAO CHEN
336	3668575	XUN ZHAO
337	4426932	YAJUAN SI
338	3800708	YALI WANG
339	4721620	YAN CHEN
340	3457484	YANBING ZHENG
341	4285354	YANG ZHAO
342	4172260	YANNAN QIU
343	4895244	YAOGUO XIE
344	4576059	YAOTAO XU
345	4695492	YAZHEN WANG
346	4285538	YI CHAI
347	5327139	YI LI
348	4586128	YI LIU
349	5693277	YIFAN MEI
350	5323691	YILIN ZHANG
351	5170492	YING ZHANG
352	5696436	YONGFENG WU
353	5644228	YONGJOON KIM
354	5518378	YONGSU LEE
355	5514517	YOUNG LEE
356	5177493	YOUNG MIN PARK
357	4278664	YOUNGDEOK HWANG
358	5326059	YOURAN QI
359	3010828	YU MENGANG
360	3961448	YU QIU LIU
361	3791787	YUAN JIANG
362	5167973	YUAN LI
363	4733966	YUAN WANG
364	4906458	YUANZHI LI
365	5903922	YUCHANG WU
366	5700058	YUCHEN ZHOU
367	4144479	YUJIN CHUNG
368	5907599	YUNONG LIN
369	5214858	YUQING XU
370	1426304	ZHANG CHUNMING
371	4108525	ZHENGJUN ZHANG
372	3107260	ZHENGXIAO WU
373	4588525	ZHIGENG GENG
374	3339239	ZHIGUO XIAO
375	5693629	ZHONGJIE YU
376	3626007	ZHUANG WU
377	5099221	ZIFENG ZHAO

378 5591917
379 4138542

ZIJIAN NI
ZUOFENG SHANG

```
dbGetQuery(conn, "SELECT COUNT(DISTINCT
    instructors.id) AS total_teachers
FROM subjects
JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code
JOIN course_offerings
    ON subject_memberships.course_offering_uuid = course_offerings.uuid
JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid
JOIN teachings
    ON sections.uuid = teachings.section_uuid
JOIN instructors
    ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
ORDER BY instructors.name;")
```

```
total_teachers
1              379
```

3. O GPA americano é definido numa escala de 0 a 4, em que $A = 4$, $AB = 3.5$, $B = 3$, $BC = 2.5$, $C = 2$, $D = 1$ e $F = 0$. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):
- Quem é o professor mais difícil?

```
dbGetQuery(conn, "SELECT
    instructors.name,
    SUM(
        gd.a_count * 4.0 +
        gd.ab_count * 3.5 +
        gd.b_count * 3.0 +
        gd.bc_count * 2.5 +
        gd.c_count * 2.0 +
        gd.d_count * 1.0 +
        gd.f_count * 0.0
    ) * 1.0 /
    SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f
    SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f
```

```

FROM subjects s
JOIN subject_memberships sm
    ON s.code = sm.subject_code
JOIN course_offerings co
    ON sm.course_offering_uuid = co.uuid
JOIN sections sec
    ON co.uuid = sec.course_offering_uuid
JOIN teachings t
    ON sec.uuid = t.section_uuid
JOIN instructors
    ON t.instructor_id = instructors.id
JOIN grade_distributions gd
    ON sec.course_offering_uuid = gd.course_offering_uuid
    AND sec.number = gd.section_number
WHERE s.abbreviation = 'STAT'
GROUP BY instructors.id, instructors.name
HAVING total_alunos > 0
ORDER BY media_gpa ASC
LIMIT 1")

```

	name	media_gpa	total_alunos
1	JAMES D KUELBS	2.6375	40

- Quem é o professor mais fácil?

```

dbGetQuery(conn, "SELECT
    instructors.name,
    SUM(
        gd.a_count * 4.0 +
        gd.ab_count * 3.5 +
        gd.b_count * 3.0 +
        gd.bc_count * 2.5 +
        gd.c_count * 2.0 +
        gd.d_count * 1.0 +
        gd.f_count * 0.0
    ) * 1.0 /
    SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f.
    SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f.
FROM subjects s
JOIN subject_memberships sm
    ON s.code = sm.subject_code
JOIN course_offerings co

```

```

    ON sm.course_offering_uuid = co.uuid
JOIN sections sec
    ON co.uuid = sec.course_offering_uuid
JOIN teachings t
    ON sec.uuid = t.section_uuid
JOIN instructors
    ON t.instructor_id = instructors.id
JOIN grade_distributions gd
    ON sec.course_offering_uuid = gd.course_offering_uuid
    AND sec.number = gd.section_number
WHERE s.abbreviation = 'STAT'
GROUP BY instructors.id, instructors.name
HAVING total_alunos > 0
ORDER BY media_gpa DESC
LIMIT 1")

```

	name	media_gpa	total_alunos
1	SRIKANTHMADHAVAN ARAVAMUTHAN	4	16

- Qual é a disciplina mais difícil?

```

dbGetQuery(conn, "SELECT
    c.name AS disciplina,
    c.number AS codigo_disciplina,
    SUM(
        gd.a_count * 4.0 +
        gd.ab_count * 3.5 +
        gd.b_count * 3.0 +
        gd.bc_count * 2.5 +
        gd.c_count * 2.0 +
        gd.d_count * 1.0 +
        gd.f_count * 0.0
    ) * 1.0 /
    SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f
    SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f
FROM subjects s
JOIN subject_memberships sm
    ON s.code = sm.subject_code
JOIN course_offerings co
    ON sm.course_offering_uuid = co.uuid
JOIN courses c
    ON co.course_uuid = c.uuid

```

```

JOIN grade_distributions gd
  ON co.uuid = gd.course_offering_uuid
WHERE s.abbreviation = 'STAT'
GROUP BY c.uuid, c.name, c.number
HAVING total_alunos > 0
ORDER BY media_gpa ASC
LIMIT 1")

```

	disciplina	codigo_disciplina	media_gpa
1	Introduction to the Theory of Probability	431	2.886374
	total_alunos		
1		2231	

- Qual é a disciplina mais fácil?

```

dbGetQuery(conn, "SELECT
  c.name AS disciplina,
  c.number AS codigo_disciplina,
  SUM(
    gd.a_count * 4.0 +
    gd.ab_count * 3.5 +
    gd.b_count * 3.0 +
    gd.bc_count * 2.5 +
    gd.c_count * 2.0 +
    gd.d_count * 1.0 +
    gd.f_count * 0.0
  ) * 1.0 /
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count)
FROM subjects s
JOIN subject_memberships sm
  ON s.code = sm.subject_code
JOIN course_offerings co
  ON sm.course_offering_uuid = co.uuid
JOIN courses c
  ON co.course_uuid = c.uuid
JOIN grade_distributions gd
  ON co.uuid = gd.course_offering_uuid
WHERE s.abbreviation = 'STAT'
GROUP BY c.uuid, c.name, c.number
HAVING total_alunos > 0

```

```
ORDER BY media_gpa DESC
LIMIT 1")
```

	disciplina	codigo_disciplina
1	Nonparametric Statistics and Machine Learning Methods	841
	media_gpa	total_alunos
1	4	24

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-22 23:41:11 -03"
```